

Lecture 14

Last time: Graph ^{simple} _{multigraph.}, degree, Ramsey, path, connected.

I Subgraphs, Connected Comp.

II k-coloring

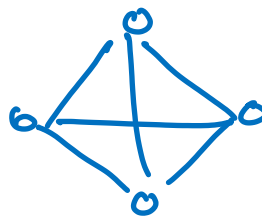
III 2-coloring

Defn: A graph $H = (W, F)$ is a subgraph of $G = (V, E)$ if $W \subseteq V$ and $F \subseteq E$.

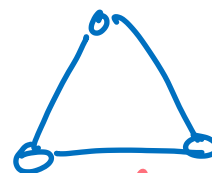
This is denoted $H \subseteq G$. If $H \subseteq G$ and $H \neq G$ then $H \subset G$. ★

ex: For $n \geq 1$, define $K_n = (V, E)$ $V = \{0, \dots, n-1\}$
 $E = \{S : S \subseteq V, |S| = 2\}$.

$n=4$



$n=3$



$$C_n = (V, F)$$



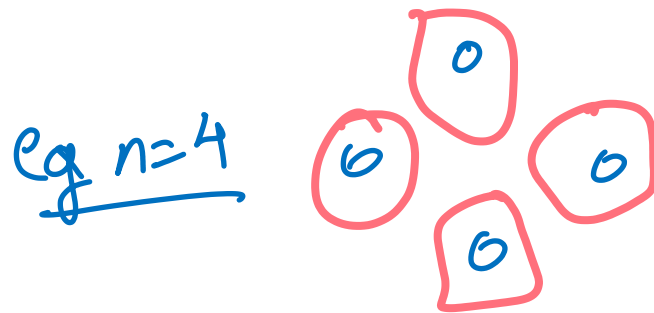
$$V = \{0, \dots, n-1\}$$

$$F = \{xy : x - y \equiv 1 \pmod{n}\}.$$

"mutual acquaint."

$$\forall n, \quad C_n \subseteq K_n. \quad F \subseteq E.$$

ex: For $n \geq 1$ define $I_n = (V, \emptyset)$ where $V = \{0, \dots, n-1\}$.



"mutual strangers"

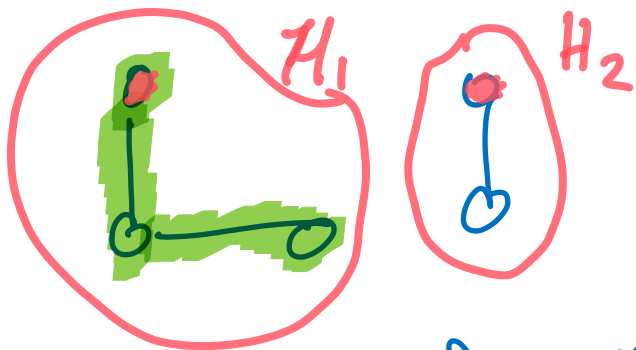
$$I_n \subseteq C_n.$$

Defn: A connected component of $G = (V, E)$ is a nonempty subgraph $H \subseteq G$ which is

- ① connected.
- ② maximal : $\forall H' \subseteq G$ ($H' \supsetneq H \rightarrow H'$ is disconnected).

Empty Graph
($\emptyset, \emptyset$)

ex:



2 connected components.

ex: If G is connected,
it is a connected comp.

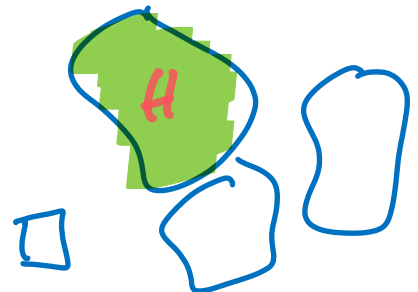
Theorem 14.1: Every graph is a disjoint union of its connected components.

Lemma: If $G = (V, E)$ is a nonempty graph, then G has at least one connected component.

Proof Sketch of Lemma: Let G be nonempty.

By the Lemma, can choose $H \subseteq G$ s.t. $H \neq \emptyset$
and H is a connected component.

Remove H , use strong induction.
< HW >



Proof of Lemma: Let $G = (V, E)$ be nonempty.

Choose a vertex $x \in G$.

Define

$$W = \{ y \in V : \text{there is a path from } x \text{ to } y \text{ in } G \} \cup \{x\}.$$

Consider the subgraph

$$H = (W, F) \text{ where}$$

$$F = \{ y_1 y_2 \in E : y_1, y_2 \in W \}.$$

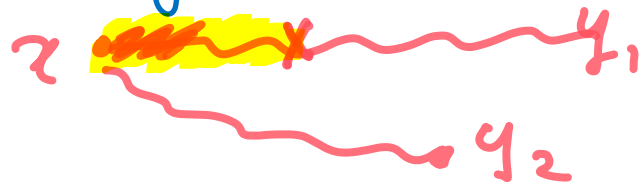
"induced subgraph"

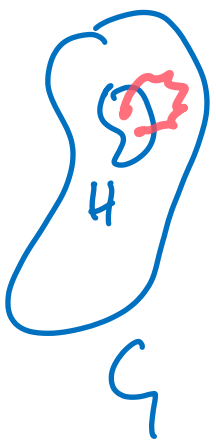
We first show H is connected.

Let $y_1, y_2 \in W$ be two distinct vertices in H .

By definition, we can choose paths γ_1 from x to y_1

and γ_2 from x to y_2 in G .

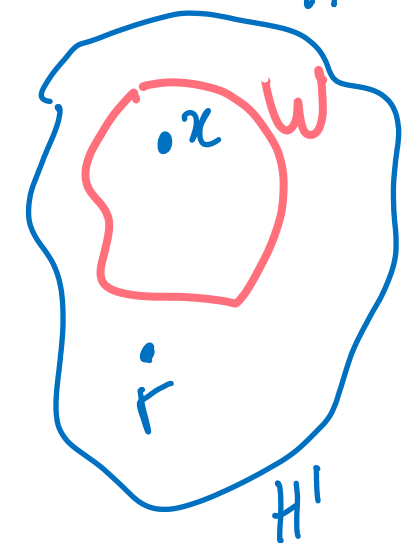




Since every vertex in γ_1, γ_2 is connected to x in G by an initial segment of the ^{corresponding} paths, γ_1 and γ_2 are paths in H . (uses the defn of H)

Let γ be the concatenation of (the reversal of γ_1) and γ_2 . Notice γ is a path from y_1 to y_2 in H .

To see that H is maximal, assume $H' \subseteq G$ and $H' \neq H$. ~~***~~ Choose a vertex $r \in H'$ s.t. $r \notin W$.



By construction, there is no path in G from x to r , so H' is disconnected, as desired. $\square$

***: Claim: H' must contain a vertex r s.t. $r \notin W$.

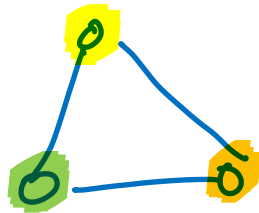
Proof: If not, then $H' = (W, F')$ for some $F' \subseteq E$. But then $F' = F$ by construction, so $H = H'$. $\square$

II k-coloring (All graphs are simple)

Defn: A k -coloring of a graph $G = (V, E)$ is a function $f: V \rightarrow \{1, \dots, k\}$ such that

$$\forall xy \in E \quad f(x) \neq f(y).$$

ex:



3-coloring.

$$\chi(K_3) = 3.$$

If a k -coloring of G exists, it is called k -colorable.

The smallest k such that G is k -colorable is called its chromatic number $\chi(G)$. Notice: $\chi(G) \leq |V|$.

ex:

$$\chi(\circ) = 1.$$

$$\chi(I_n) = 1 \quad \forall n \geq 1.$$

Theorem 14.2 : If $G=(V,E)$ has maximum degree D ,
then $\chi(G) \leq D+1$.

Application : 40k at Cal.

4,000 upper div courses at Cal, Each Course has ≤ 50 students.
Every student takes ≤ 2 upper division courses.

(Q) How many final exam slots are needed?

(A) Define a conflict graph :

G $V = \{c : c \text{ is an upper div course}\}$. $|V|=4000$.
 $E = \{c_1, c_2 : \text{there is a student in both } c_1 \text{ and } c_2\}$.

Observe : $\forall c \in V$, $\deg(c) \leq 50$ since each student
takes ≤ 1 other course.

By thm, G is 51-colorable, i.e. $\exists f: V \rightarrow \{1 \dots 51\}$
s.t. $\forall c_1, c_2 \in E \quad f(c_1) \neq f(c_2)$.

Proof: Let $D \geq 0$. We proceed by induction.

Let $P(n) = "$ $\forall G$ with $\boxed{\leq n}$ vertices,
($\max \deg(G) \leq D \rightarrow \chi(G) \leq D+1$) $"$.

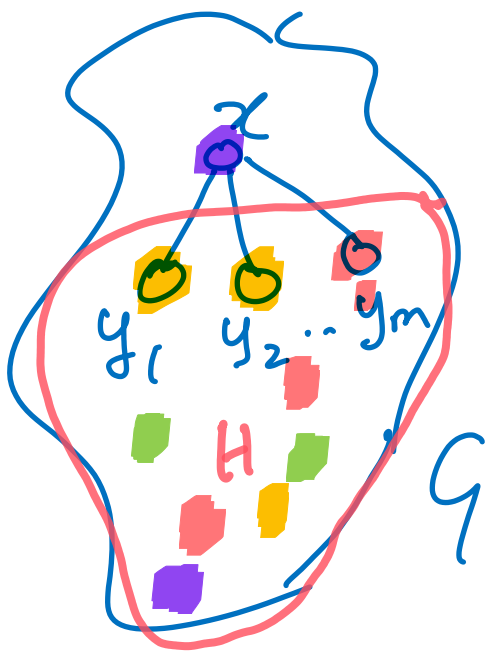
Base Case: If G has 1 vertex, it is 1-colorable
so $\chi(G) = 1 \leq D+1$. ○

Inductive Step: Assume $k \in \mathbb{Z}_+$. Assume $P(k)$.

We will show $P(k+1)$.

Assume $G = (V, E)$ has $k+1$ vertices
and maximum degree at most D .

G



Choose a vertex $x \in V$.

Let $y_1, y_2, \dots, y_m$ be the vertices adjacent to x in G .

Let $H = (W, F)$ be the subgraph obtained by deleting x :

$$W = V - \{x\}$$

$$F = \{yz \in E : y, z \in W\}.$$

Since H has $\max \text{degree} \leq D$ and k vertices,

$P(k)$ implies that there is a $D+1$ -coloring

$$f: W \rightarrow \{1, \dots, D+1\} \text{ of } H.$$

Observe that since $m < D+1$, there is at least one color $c \in \{1, \dots, D+1\}$ s.t. $c \neq f(y_1), f(y_2), \dots, f(y_m)$

Define $g: V \rightarrow \{1, \dots, D+1\}$ by

$$\begin{cases} g(y) = f(y) & \text{if } y \neq x \\ g(x) = c. \end{cases}$$

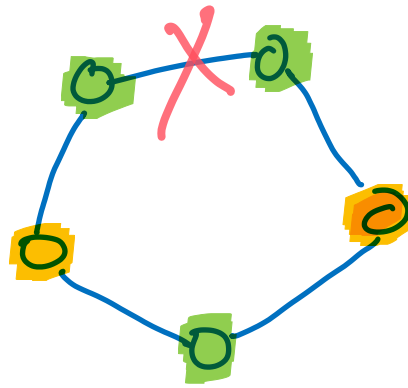
g is a valid coloring of H and also
of the edges $xy_1, xy_2, \dots, xy_m$.

So G is $D+1$ colorable, as desired. $\square$

Lessons:

- Name paths, vertices, subgraphs.
- Concatenation, reversal.
- paths in G vs. paths in H .
- Deleting a vertex yields a subgraph.

Poll:



Not 2-colorable.

